INCH-POUND
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August 8 2022

COMMERCIAL ITEM DESCRIPTION

MANUAL WELDING MACHINE, GAS TUNGSTEN ARC WELDING (GTAW), AC/DC INVERTER, WATER COOLED, CONSTANT CURRENT

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers a constant alternating current/direct current (AC/DC) inverter for Gas Tungsten Arc Welding (GTAW) welding machines. GTAW, often referred to as Tungsten Inert Gas (TIG) welding, is a welding process using an arc between a tungsten electrode (non-filler, non-consumable) and the weld pool. The process is used with a shielding gas and without the application of pressure. The equipment described herein is for Air Force (AF) Metals Technology (MT) shop installations. AF MT shops serve as the organic and local manufacturing / fabrication resource across the enterprise (150+ locations).

1.1 Intended use. The welding machines covered by this CID are intended for manual welding used in operations that require the efficient and permanent joining of similar, and in some cases, dissimilar metal pieces. The machines are intended for use in Air Force Metals Technology shops via Air Force Specialty Code (AFSC) 27AX1.

2. CLASSIFICATION. The welding machines shall be of the following types per Table 1.

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TABLE I. Types of GTAW Welding Machines.

Type	Rated to weld material thicknesses (inches)	Material	Input power (volts 1- or 3-phase)	Input (Hz)	Min/Max amps (A)	Portable cart and cooler	Indoor and/or outdoor use
01	0.015-3/8	Aluminum	208/240/480	50/60	5/300	Required	Indoor
	0.012-1/2	Steel					
02	0.015-1/2	Aluminum	208/240/481	50/60	5/400	Required	Indoor
	0.012-5/8	Steel					
03	0.012-3/8	Aluminum	208-575	50/60	1/280	Required	Indoor & outdoor
	0.004-3/8	Steel					
04	0.015-5/8	Aluminum	208-575	50/60	3/400	Required	Indoor & outdoor
	0.012-5/8	Steel					

3. SALIENT CHARACTERISTICS.

3.1 General requirements. The welding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine shall be designed and manufactured in accordance with the American National Standards Institute/National Electrical Manufacturers Association (ANSI/NEMA) EW-1.

3.2 Components. GTAW requires the use of a constant-current power source, an inert gas, gas-regulating equipment, a torch, heavy-duty ground clamps (spring style) and filler metal appropriate for the machine rating. Additionally, the welding machine shall include, as a minimum, the following components.

3.2.1 Enclosure. The welding-machine enclosure shall be fan (main/standard unit) and water (interconnected option) cooled and manufactured to withstand the rough handling normally encountered in construction and industrial/machine shop areas. The enclosure shall be constructed of metal materials (sheet metal, cast, wrought), and the design shall be resistance to impact, combustibility, temperature distortion, corrosion, and thermal aging experienced during normal service use and within non-environmentally/climatically controlled machine-shop locations at various Air Force locations. Ventilation holes shall be covered with screens or shall be small enough to prevent the entry of debris. The enclosure shall be fitted with handles, wheels, and lifting means/eyes. All electrical connection points and machine controls shall be clearly labeled.

3.2.2 Power source. The welding-machine power source shall be the inverter type capable of delivering the required AC or DC current output and shall maintain constant current during the welding process. The power source shall have thermal overload protection and power-factor correction. A foot pedal shall have the options to i) control welding current during the welding process or ii) serve as a current on/off switch. If the foot pedal is set up to serve as an on/off switch, then the amperage is set at the welding machine by the user and cannot be varied during the welding process; that is, the pedal serves as on (current flowing) or off (current not flowing). If the foot pedal is set up to control and to allow the variance of amperage, then pressing the pedal results in increased welding current. In this capacity, the weld pedal affords the user the ability to vary the welding amperage from a very low level (low-level pedal compression) up to the amperage level set on the welding machine (full-pedal compression).

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3.2.3 High-frequency generator. Tungsten electrodes can become contaminated when touched to the work piece during the arc-initiation process. To prevent this, the welding machine shall employ a high-frequency voltage that is capable of jumping the arc gap between the tungsten electrode and the work piece in order to serve as a welding starting circuit (no physical contact required). Thus, the welding machine shall be equipped with a high-frequency (HF) generator for DC arc initiation and AC arc stabilization.

3.2.4 Controls. The minimum welding machine controls for Type 01 and Type 02 welder machines are described in Table II.

TABLE II. Controls and display for Type 01 and Type 02 welder machines.

Control	Display/Function
Ammeter.	Displays actual amperage while welding and preset amperage while idle. Used to display parameter selection options.
Voltmeter.	Displays actual rectified average voltage when voltage is present at the weld output terminals. Used to display parameter description(s).
Main power switch.	Used to turn the machine on and off.
Output ON indicator.	Indicator (non-white light) illuminates when output is on.
Processor selector.	AC TIG - Used for welding aluminum.
	DC TIG (DCEN) - Used for welding mild and stainless steels.
	DC TIG pulse (direct-current electrode negative - DCEN) - used for welding mild and stainless steels.
	DC stick (direct current electrode positive - DCEP) - used for welding steels.
Memory card port and indicator.	Used to add features to the machine and update software to the boards within the machine. Indicator is lit while the card is being accessed.
Amperage adjustment control.	Used to change preset amperage value.
Menu buttons.	Buttons, or similar style, used to scroll through available parameters for the selection process.
Lock parameters function.	Locks in all parameters to factor settings; must have a means to reset to factory settings (example: holding 3-5 seconds to reset all parameters to factory settings. Meter display shall show countdown).
Error log.	Records error events.
Sleep timer.	1/5/10/15/30/45/60 minutes.
Cooler power	Auto/On/Off.
External pulse Control.	On/Off.
Weld timers.	On/Off.
Arc time.	0–9,999 hours, 0–59 minutes, 0–999,999 cycles, resettable.

Note: The minimum welding machine controls for Type 03 and Type 04 welder machines are described in Table III.

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TABLE III. Controls and display for Type 03 and Type 04 welder machines.

Control	Display/Function	
Ammeter.	Displays actual amperage while welding and preset amperage while idle. Used to display parameter selection options.	
Voltmeter.	Displays actual rectified-average voltage when voltage is present at the weld-output terminals. Used to display parameter description(s).	
Polarity	AC/DC.	
Process / arc Starting.	TIG: High-frequency impulse, lift-arc.	
	Stick: Adaptive hot start (a very brief current pulse at arc initiation to stabilize the arc quickly).	
Output control.	Standard remote, 2T trigger hold, output on.	
Pulse control	Pulses per second:	DC: 0.1–500 pps
		AC: 0.1–500 pps
	Peak time.	5–95%
	Background amps.	5–95%
Memory card port and indicator.	Used to add features to the machine and update software to the boards within the machine. Indicator is lit while the card is being accessed.	
Activity indicator.	Illumination showing on/activity.	
Encoder control.	Knob style.	
AC Wave shape.	Balance.	50–99% EN
	Frequency.	20–400 HZ
Gas/DIG.	Pre flow.	0.0–25.0 Seconds
	Post flow.	Auto/off–50 seconds
	DIG.	Off–100%
Error log.	Records error events.	
Sleep timer.	1/5/10/15/30/45/60 minutes.	
Cooler power.	Auto/on/off.	
External pulse control.	On/off.	
Weld timers.	On/off.	
Arc time.	0–9,999 hours, 0–59 minutes, 0–999,999 cycles, resettable.	

3.2.5 Gas shielding and cooling water. The welding machine shall provide a means to dispense and adjust the flow of shielding gas and shall come standard water cooled. The shielding-gas and cooling-water controls and metering components shall be contained on the welding machine. The welding machine shall be capable of utilizing coolants containing a base of ethylene glycol and deionized water to protect against freezing or boiling temperatures. In addition, the welding machine shall be capable of utilizing coolants that contain compounds that resist algae growth.

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TABLE IV. Cooler Specifications.

Input Power	Max Current Draw	Max Cooling Capacity (W)	Rated Cooling Capacity (W)	Tank Capacity (gal)	Max Dimensions (HxWxD inches)	Net Weight (lbs)
115V, 60 Hz	4.7A (60 Hz)	3400	1300	1.25-1.5	11.5x10.5x25	40-45

3.3 Electrical system. See Table1 for welding-machine specifications and Table IV for water cooler specifications.

3.4 Performance. The welding machine shall meet the performance requirements of ISO 1708 when tested in accordance with sections 5.3 through 5.3.4.

3.5 Portability. The welding machines shall be portable via roller cart, and shall be base mounted. The cart shall have four wheels (front two wheels shall swivel; rear wheels shall not swivel), shall be able to be locked into position by mechanical/brake means, and shall support at least one single-cylinder rack, a foot-pedal holder, two cable/torch holders, and two TIG filler-rod holders.

3.6 Safety and health requirements. The manufacturer shall ensure that the welding machine and all equipment and accessories used on the machine shall be in compliance with American Welding Society (ANSI/AWS) Z49.1 and Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI/AWS and OSHA standards, the OSHA standards shall apply. Covers, guards, or other safety devices shall be provided for all parts of the system that present safety hazards. The manufacturer shall ensure that the machine is intended for manual use. A machine having a power source with an open-circuit voltage greater than those listed in NEMA EW-1 section 5.3.2 shall not be provided as a manual system.

3.7 System of units. The U.S. Customary System of Units (US) shall be used to graduate measuring and indicating devices.

3.8 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified, the nameplate shall contain the following information:

a. Manufacturer's name.
b. Machine model designation.
c. Power input (volts, total amperes, phase, and frequency).
d. Rated load volts and amperes.
e. Maximum open circuit voltage.
f. Duty cycle at rated load.
g. Contract number or order number.
h. National stock number or plant equipment code.
i. Date of manufacture.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to

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during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality-assurance practices; and be the same system offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Inspection. The arc-welding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified, the welding machine shall be tested in accordance with sections 5.3.1 through 5.3.3.

5.3.1 Test setup and conditions. The manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certificationof calibration relevant to each test.

5.3.2 Operational test. The arc-welding machine shall be operated in accordance with the manufacturer's standard operating procedures. Proper operation of all controls and features shall be verified during the test period.

5.3.3 Performance tests. The performance tests shall be conducted in accordance with the following sections of NEMA EW-1:

a. Drop test - section 3.8.2 of NEMA EW-1.
b. Output test - section 5 of NEMA EW-1.
c. Temperature rise test - section 6 of NEMA EW-1.
d. High-potential test - section 7 of NEMA EW-1.
e. High-frequency generator test - section 10 of NEMA EW-1.

5.4 Acceptance. Preliminary and final acceptance tests shall be conducted at the manufacturer's site or at a location agreed upon by the Air Force (AF) Metals Technology Office / Metals Fabrication and Technologies Office (MTO/MFTO), Robins AFB, GA. Failure of the machine to meet the tests called out in section 5.3 through 5.3.3 shall be cause for rejection. AF MTO/MFTO are the engineering review and approval authority for all testing resulting from the requirements listed herein.

6. PACKAGING.

6.1 Packaging requirements. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Government documents. Copies of specifications, standards, and handbooks may be obtained at https://assist.dla.mil

7.1.2 American National Standards Institute (ANSI) documents. Copies of ANSI documents may be obtained at https://ansi.org/.

7.1.3 National Electrical Manufacturers Association (NEMA) Copies of NEMA documents may be obtained at https://www.nema.org/.

7.1.4 Occupational Safety and Health Association (OSHA). Copies of OSHA documents may be

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obtained at https://www.osha.gov/.

7.1.5 Environmental Protection Agency (EPA). Copies of EPA documents may be obtained at https://www.epa.gov/.

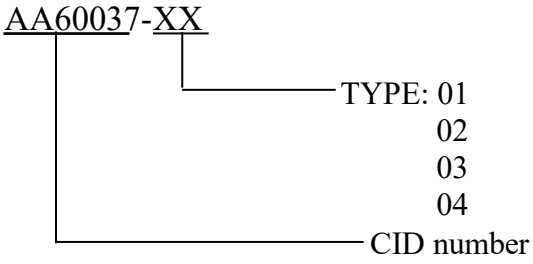
7.1.6 Federal Acquisition Regulation (FAR). Copies of FARS may be obtained at https://www.acquisition.gov/.

7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.

b. Type of arc-welding system required (Table I).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.



MILITARY INTERESTS:

Custodian:
Air Force – 184

Preparing Activity:
Air Force – 184

Reviewer:
GSA FAS

Agent:
Air Force – 110

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